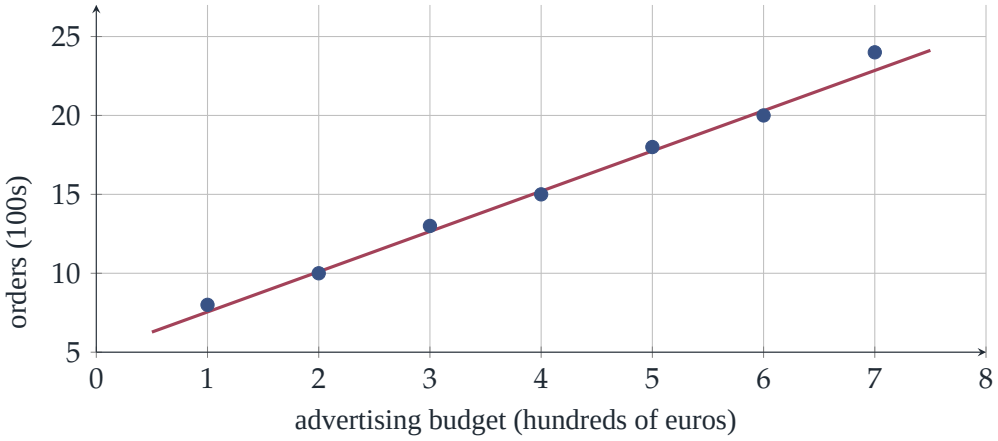
Chapter 5

Statistics with two quantitative variables

Skills: represent, choose a fit, interpolate, extrapolate with caution, communicate

5.1 Reference notes

A two-variable statistical series consists of pairs (x_i, y_i). The scatter plot allows to observe a trend, a spread and any outliers.



Key ideas

A line of best fit $y = ax + b$ provides estimates, not exact equalities. The coefficient a describes the estimated average change in y when x increases by one unit. An interpolation remains within the range of observed values; an extrapolation goes outside this interval and must be presented with caution.

The mean point $G(\bar{x}, \bar{y})$ has coordinates $\bar{x} = \dfrac{x_1 + \cdots + x_n}{n}$ and $\bar{y} = \dfrac{y_1 + \cdots + y_n}{n}$. A line of regression obtained on the calculator passes through G, up to rounding.

Method

To make use of $y = ax + b$: 1) replace x by the required value to estimate y ; 2) solve $ax + b = y_0$ to estimate x ; 3) round according to the context; 4) state interpolation or extrapolation ; 5) recall that the result depends on the model.

A change of variable, for example $z = \log(y)$ or $z = \dfrac{1}{y}$, can make a scatter plot approximately ly aligned. We then fit (x, z), then return to the variable y.

Session 31 — Scatter plot

Two-variable statistics

26/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Draw a scatter plot.

· Choose the scales.

· Identify the variables.

Fluency · 6 to 8 minutes, without calculator

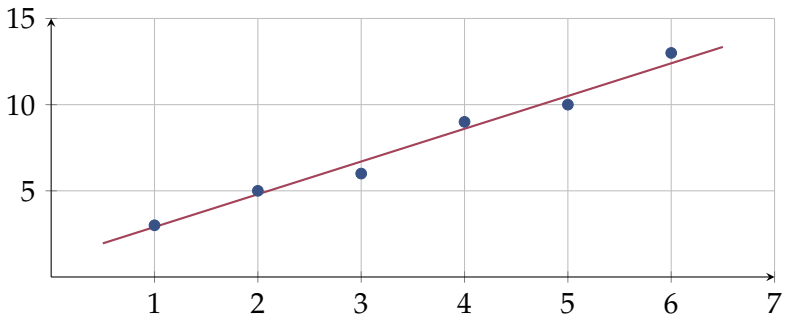
1. Write the multiplier for a 10% increase
2. Solve 3 $x + 4 = 25$. .....
3. Calculate the mean of 6, 8 and 10
4. Give the slope between $(1, 3)$ and $(4, 9)$. .....
5. With $y = 2x + 5$, calculate y for $x = 7$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: draw a scatter plot and choose scales.

Investigation

The table (x_i, y_i) gives the expenses advertising and thethe sales of six months : $(1, 12), (2, 15), (3, 17), (4, 22), (5, 24), (6, 28)$. Choose scales, plot the scatter diagram and place the mean point.



Written summary

A scatter graph represents the pairs (x_i, y_i). Axes, units and scales must be shown ; each pair gives a single point.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. For $y = 3x + 13$, estimate y when $x = 8$.
2. Solve 3 $x + 13 = 50$.
3. The model was drawn up on $[2, 12]$: comment on the estimates for $x = 8$ and $x = 30$.

References in the textbook

Extensions: Calao, courses p. 110·113; exercises p. 114, n° 1 to 6.
session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 114, Exercise 2

Plot the scatter graph of the following statistical series in a coordinate system.

x_i	-3	0	3	5	8
y_i	1	-1	-2	5	7

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 275 and the rate or common ratio by 3. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 32 — Reading and interpreting the scatter plot

Two-variable statistics

28/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Describe a trend.

· Identify spread and outliers.

· Avoid unjustified causal claims.

Fluency · 6 to 8 minutes, without calculator

1. Simplify 3 $x + 7 - x + 2$. .....
2. Calculate the mean of 7, 9 and 11
3. Give the slope between $(1, 3)$ and $(4, 9)$. .....
4. With $y = 2x + 5$, calculate y for $x = 7$. .....
5. Solve 3 $x + 4 = 25$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: describe a trend and identify spread and outliers.

Investigation

Compare three scatter plots: tight increasing trend, spread-out decreasing trend, no trend. Write two sentences for each without using the word "cause".

Written summary

Describing a scatter plot means commenting on direction, shape, spread and outliers, without asserting a causal link that has not been established.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. For $y = 4x + 14$, estimate y when $x = 8$.
2. Solve 4 $x + 14 = 50$.
3. The model was drawn up on $[2, 12]$: comment on the estimates for $x = 8$ and $x = 30$.

References in the textbook

Extensions: Calao, p. 116, nos. 11 to 15. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the

digital resources of the textbook.

Calao textbook · p. 114, Exercise 6

The graph in the textbook shows the number of cigarettes sold in France as a function of the number of wolf packs recorded in France between 1995 and 2017. How can the trend in cigarette sales be described from this graph? For even so, can we think there is a link between these facts?

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 280 and the rate or common ratio by 4. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 33 — Linear fit

Two-variable statistics

02/02/2027 • B : 2 h – C : 1 h

Objectives of the session

· Obtain an affine fit.

· Interpret a and b.

· Check with the mean point.

Fluency · 6 to 8 minutes, without calculator

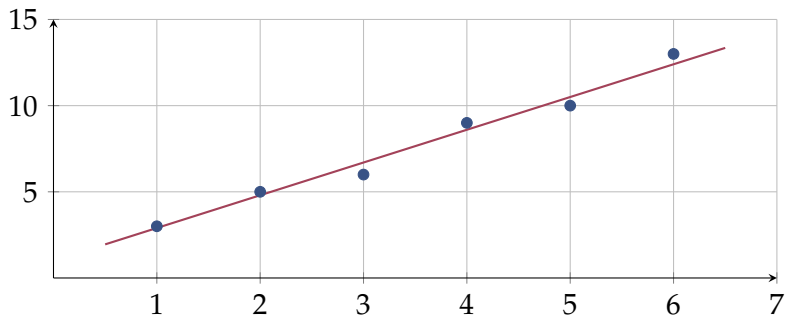
1. Calculate 20 per cent of 160 .....
2. Give the slope between $(1,3)$ and $(4,9)$. .....
3. With $y = 2x + 5$, calculate y for $x = 7$. .....
4. Solve 3 $x + 4 = 25$. .....
5. Calculate the mean of 8, 10 and 12

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: obtain a linear fit and interpret a and b.

Investigation

With the data from session 31, obtain the linear regression on the calculator. Interpret the slope and check that the mean point lies approximately on the line.



Written summary

A line $y = ax + b$ fits a scatter graph. a represents the estimated average change in y for a additional unit of x.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. For $y = 5x + 15$, estimate y when $x = 8$.
2. Solve 5 $x + 15 = 50$.
3. The model was drawn up on $[2, 12]$: comment on the estimates for $x = 8$ and $x = 30$.

References in the textbook

Extensions: Calao, courses p. 110·113; exercises p. 117, n° 16 to 18.
session is identified below by its page and number; the reference also makes it possible to find
the figures and the textbook's digital resources.

Calao textbook · p. 115, Exercise 9

1. Draw the scatter plot of the statistical series shown, in a coordinate system.

x_i	-2	-1	0	2	3	5
y_i	4	8	12	16	16	20

Draw the line with equation $y = 2x + 10$ in this coordinate system. Fitting the scatter plot with this line seem satisfactory? If not, suggest another one, giving its equation
reduced form.

2. Using this equation, determine:

· the y-coordinate of the point on the line of best fit with x-coordinate 1;

· the y-coordinate of the point on the line of best fit with x-coordinate 10;

· the x-coordinate of the point on the line of best fit with y-coordinate 7.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with
285 and the rate or common ratio by 5. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and
give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 34 — Interpolation

Two-variable statistics

04/02/2027 • B : 1 h – C : 2 h

Objectives of the session

· Interpolate.

· Solve with the model.

· Express an estimate.

Fluency · 6 to 8 minutes, without calculator

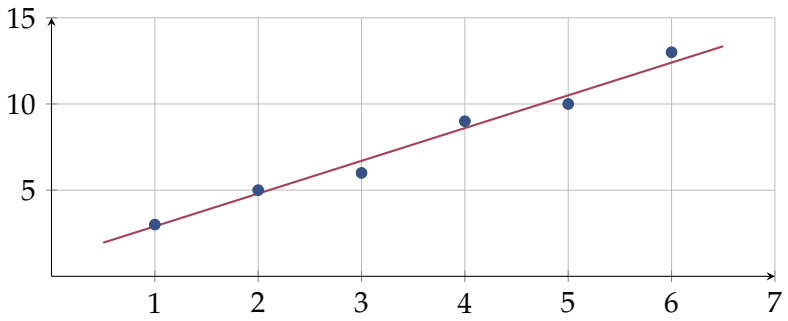
1. Write the multiplier for a 25% increase
2. With $y = 2x + 5$, calculate y for $x = 7$. .....
3. Solve 3 $x + 4 = 25$. .....
4. Calculate the mean of 9, 11 and 13
5. Give the slope between $(1, 3)$ and $(4, 9)$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: interpolate and solve with the model.

Investigation

Over the period observed $x \in [1, 12]$ the model is $y = 2{,}8x + 14$. Estimate y for $x = 7{,}5$, then estimate x when $y = 35$. Comment on each estimate.



Written summary

An interpolation estimates a value within the observed interval. The result is a model-dependent estimate.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. For $y = 6x + 16$, estimate y when $x = 8$.
2. Solve $6 \quad x + 16 = 50$.
3. The model was drawn up on $[2, 12]$: comment on the estimates for $x = 8$ and $x = 30$.

References in the textbook

Extensions: Calao, p. 118, nos. 19 to 24. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the textbook.

Calao textbook · p. 115, Exercise 8

The table below gives the number of new electric vehicles registered in France from 2014 à 2018.

Year		2014	2015	2016	2017	2018
Position in the year	x_i	0	1	2	3	4
Number of vehicles (thousands)	y_i	10,5	17,3	21,7	24,9	31

1. Draw the scatter plot of this time series in a coordinate system.
2. Propose a linear fit for this scatter plot.
3. Assuming this fitted model remains valid for several years, how many vehicles new cars would have been registered in 2020? From which year would the number exceed 50,000 new vehicles registered?

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 290 and the rate or common ratio by 6. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 35 — Extrapolation

Two-variable statistics

09/02/2027 • B : 2 h – C : 1 h

Objectives of the session

· Extrapolate carefully.

· Evaluate plausibility.

· Present a limitation of the model.

Fluency · 6 to 8 minutes, without calculator

1. Simplify 3 $x + 3 - x + 2$. .....

2. Solve 3 $x + 4 = 25$. .....

3. Calculate the mean of 3, 5 and 7

4. Give the slope between $(1, 3)$ and $(4, 9)$. .....

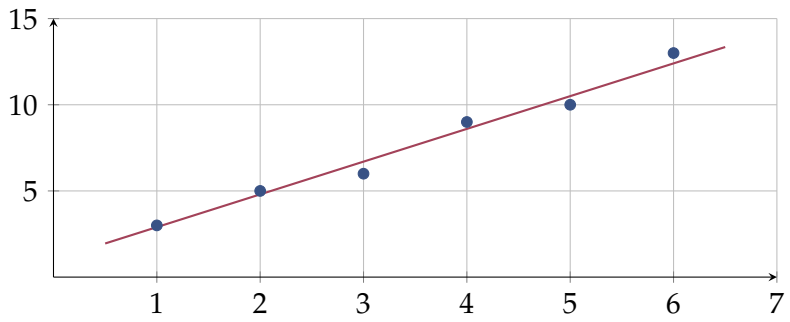
5. With $y = 2x + 5$, calculate y for $x = 7$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: extrapolate carefully and assess plausibility.

Investigation

The same model $y = 2{,}8x + 14$, fitted on $[1, 12]$, is used for $x = 40$. Calculate the estimate, then give three reasons why it can be unreliable.



Written summary

An extrapolation estimates outside the observed interval. The further away we go, the more caution is necessary.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. For $y = 2x + 12$, estimate y when $x = 8$.
2. Solve 2 $x + 12 = 50$.
3. The model was drawn up on $[2, 12]$: comment on the estimates for $x = 8$ and $x = 30$.

References in the textbook

Extensions: Calao, p. 119, nos. 25 to 27. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the textbook.

Calao textbook · p. 118, Exercise 20

The annual turnover of a company (in thousands of euros) was recorded over its first five first years of existence.

Year		2016	2017	2018	2019	2020
Position in the year	x_i	0	1	2	3	4
Turnover	y_i	315,6	489,9	579,3	763,5	874,5

1. Plot, in an orthogonal coordinate system, the scatter graph $M_i(x_i; y_i)$ of this statistical series.
2. This scatter plot is fitted by the line (M_1M_5). Determine an equation of this line of fit (round the coefficients to one decimal place).
3. If the trend continues, what turnover can be forecast for 2024? At the end of What year will the turnover exceed ·2 million?

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 295 and the rate or common ratio by 7. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 36 — Change of variable

Two-variable statistics

11/02/2027 • B : 1 h – C : 2 h

Objectives of the session

· Make a change of variable.

· Fit the new data.

· Return to the initial variable.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 80 .....

2. Calculate the mean of 4, 6 and 8

3. Give the slope between $(1,3)$ and $(4,9)$. .....

4. With $y = 2x + 5$, calculate y for $x = 7$. .....

5. Solve 3 $x + 4 = 25$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: change the variable and fit the new data.

Investigation

The data appear exponential. Let $z = \log(y)$ and the fit gives $z = 0{,}08x + 1{,}2$. Express y as a function of x, then calculate y for $x = 5$.

Written summary

A change of variable can straighten out a scatter graph. After fitting in the new variable, it you must return to the original variable.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. For $y = 3x + 13$, estimate y when $x = 8$.

2. Solve 3 $x + 13 = 50$.

3. The model was drawn up on $[2, 12]$: comment on the estimates for $x = 8$ and $x = 30$.

References in the textbook

Extensions: Calao, p. 120, n° 28·29. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the

digital resources of the textbook.

Calao textbook · p. 120, Exercise 29

In a concert hall, sound intensity levels from a generator of
tions was carried out at several locations. The measurements taken were recorded in the table below.

Distance x_i (m)	0,5	1	2	3	5	10	11	12	15	20
Sound intensity y_i (dB)	97	85	73	66	59	53	55	52	55	49

Since an affine fit does not seem appropriate, a change of variables is considered.

1. Complete the table from the textbook with $u_i = \log(x_i)$.
2. Plot the scatter graph $(u_i; y_i)$ in a coordinate system, then propose a linear fit.
3. Deduce an expression for the sound intensity in terms of the distance.
4. From the expression found, determine the expected sound intensity at a distance of 8 metres, then 30 metres.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 300 and the rate or common ratio by 2. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 37 — Choosing a model and fitting

Two-variable statistics

02/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Compare models.

· Use the residuals.

· Define a domain of validity.

Fluency · 6 to 8 minutes, without calculator

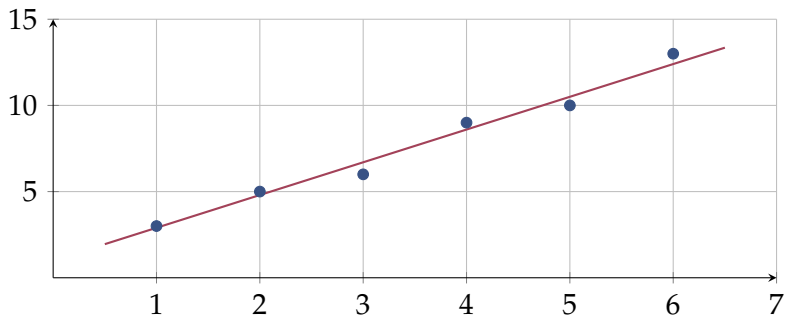
1. Write the multiplier for a 10% increase
2. Give the slope between $(1,3)$ and $(4,9)$. .....
3. With $y = 2x + 5$, calculate y for $x = 7$. .....
4. Solve 3 $x + 4 = 25$. .....
5. Calculate the mean of 5, 7 and 9

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: compare models and use residuals.

Investigation

Two models give total absolute residuals 18 ,4 and 11,7 on the observed range. The second however shows unrealistic growth after $x = 20$. Choose a model to interpolate at $x = 12$, then discuss extrapolation to $x = 50$.



Written summary

A model is chosen according to the shape of the scatter plot, the size of the residuals and the intended range of use. A best visual fit does not allow unlimited extrapolation.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. For $y = 4x + 14$, estimate y when $x = 8$.
2. Solve 4 $x + 14 = 50$.
3. The model was drawn up on $[2, 12]$: comment on the estimates for $x = 8$ and $x = 30$.

References in the textbook

Extensions: Calao, p. 121, nos. 30 to 32, then p. 122, no. 33. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 119, Exercise 25

We want to determine the caffeine concentration of a drink by measuring absorbance. For this, a laser beam is used and the attenuation of the emitted light is measured after it passes through the liquid. For the purposes of the experiment, several solutions of which the caffeine content is known are tested and the following set of values is obtained.

Absorbance A	0,155	0,320	0,465	0,630	0,935
Concentration C (mg/L)	5	10	15	20	30

1. Represent these data with a scatter plot in a coordinate system.
2. Does an affine fit seem appropriate here? If so, express the value C of the caffeine concentration as a function of value A of the absorbance.
3. For a drink containing caffeine, an absorbance of 0 is measured ,5. What then is its caffeine concentration?

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 305 and the rate or common ratio by 3. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 38 — Review and assessment

Two-variable statistics

04/03/2027 • B : 1 h – C : 2 h

Objectives of the session

· Choose and use a fit.

· Write a statistical conclusion.

· Analyse an error.

Fluency · 6 to 8 minutes, without calculator

1. Simplify 3 $x + 6 - x + 2$. .....

2. With $y = 2x + 5$, calculate y for $x = 7$. .....

3. Solve 3 $x + 4 = 25$. .....

4. Calculate the mean of 6, 8 and 10

5. Give the slope between $(1, 3)$ and $(4, 9)$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: choose and use a fit and write a statistical conclusion.

Investigation

From a scatter plot provided by the teacher, produce a complete answer: choice of model, equation, estimate, type of estimate, unit and caveat. Swap papers and use a marking grid of six criteria.

Written summary

A complete statistical answer includes the model, the calculation, the rounding, the unit, the type of estimate and a caveat about validity.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. For $y = 5x + 15$, estimate y when $x = 8$.

2. Solve 5 $x + 15 = 50$.

3. The model was drawn up on $[2, 12]$: comment on the estimates for $x = 8$ and $x = 30$.

References in the textbook

Extensions: Calao, p. 122·123. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the resources digital resources of the textbook.

Manuel Calao · p. 118·119, exercise 24

Choose the right proposal. Justify the answer. We consider the cloud of points $M_1(1;16)$, $M_2(3;13)$ and $M_3(5;4)$.

1. The mean point of this scatter plot is:

   a. the point with coordinates $(3;10)$;

   b. the point M_2 ;

   c. the point with coordinates $(3;11)$.

2. The line (M_1M_3) has equation:

   a. $y = 4x + 16$;

   b. $y = -3x + 19$;

   c. $y = -2{,}93x + 17{,}7$.

3. The regression line given by the calculator:

   a. passes through one of the three points;

   b. is parallel to the line (M_1M_3) ;

   c. goes through the origin of the coordinate system.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 310 and the rate or common ratio by 4. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.